Plug-and-Play Image Restoration: A Two-Level Coarse-to-Fine Initialization Study Using DeepInverse and DRUNet

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Abstract. This report describes a controlled experiment comparing three initialization strategies for Plug-and-Play image restoration with a pretrained DRUNet denoiser. We degraded four standard test images with synthetic Gaussian blur and noise, then restored each image five times (different noise seeds) under three conditions: standard DPIR, observation-only initialization, and two-level coarse PnP initialization. The two-level approach produced a mean PSNR of 28.593 dB compared to 28.411 dB for standard DPIR, with all 20 paired comparisons showing positive gains. The improvement comes at the cost of roughly 39% more computation time. We also compare learned DPIR against classical Total Variation restoration, examine convergence behavior, and test noise-level robustness. All code, data, and figures are publicly available for independent verification.

1. Introduction

Recovering a clean image from blurred and noisy measurements is a standard problem in computational imaging. Given an observation $y = Ax + n$, where A is a known blur operator and n is Gaussian noise, the goal is to find the best estimate of x. Variational methods approach this by minimizing a data-fidelity term plus a regularizer that encodes prior knowledge about natural images.

Plug-and-Play (PnP) methods replace the explicit regularization step with a call to a pretrained denoiser. The idea, introduced by Venkatakrishnan et al. [3] in 2013, is that a good denoiser implicitly encodes a prior over clean images. When paired with a half-quadratic splitting (HQS) optimization loop, this approach is both flexible and effective. Zhang et al. [2] demonstrated this concretely with DPIR, using a deep residual U-Net denoiser (DRUNet) that achieves competitive or superior results across deblurring, super-resolution, and inpainting tasks without any task-specific training.

A natural question arises about initialization: standard DPIR starts the HQS iterations from the raw observation y. Could we do better by first solving a cheaper, lower-resolution version of the problem and using that estimate as a warm start? This idea has roots in multigrid methods from numerical analysis, where coarse-level corrections accelerate convergence of fine-level solvers.

In this study, we tested exactly that. We compared three initialization strategies on four standard test images, each evaluated across five independent noise realizations, yielding 60 total reconstruction experiments. The work was conducted in Google Colab using the open-source DeepInverse library [1] and pretrained DRUNet weights.

2. Problem Setup and Degradation Model

We selected four color images from scikit-image's standard collection: Astronaut (a portrait with fabric and background detail), Chelsea (a tabby cat with fine fur texture), Coffee (a cup with smooth gradients and flat surfaces), and Rocket (a launch vehicle against sky, with strong edges). Each was resized to 128 x 128 pixels and normalized to the [0, 1] intensity range.

The forward degradation model applied a 2D Gaussian blur kernel (size 7 x 7, standard deviation 2.0) followed by additive white Gaussian noise with standard deviation 0.03. This produces a moderately degraded observation that is visually recognizable but substantially softened and noisy. The degradation was applied identically to all four images, varying only the random seed for the noise component. Seeds 0 through 4 were used, giving five observations per image and 20 total degradation instances.

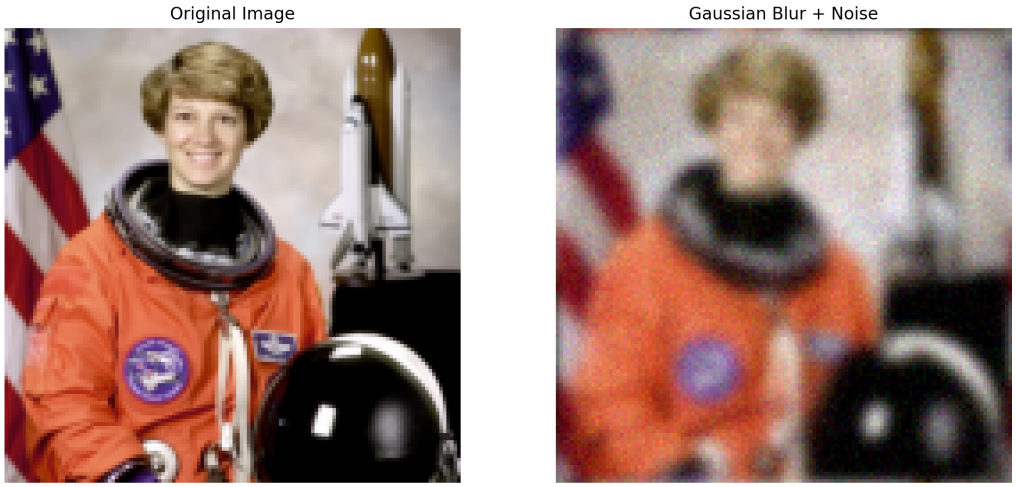


Figure 1. Degradation pipeline: ground truth, point-spread function, frequency spectrum showing high-frequency attenuation, and resulting observation.

3. Reconstruction Methods

All three methods used the same DRUNet denoiser (pretrained weights from the DeepInverse model repository, no fine-tuning) within an HQS optimization loop. The only difference between them is how the initial estimate x_0 is constructed before iterating.

3.1 Direct DPIR

The baseline. The HQS solver is initialized directly with the corrupted observation: x_0 = y. The solver then alternates between a data-fidelity gradient step and a DRUNet denoising step at progressively decreasing noise levels. This is the standard approach described by Zhang et al. [2].

3.2 Observation initialization

A control experiment. The observation y is first downsampled to 64 x 64 using bicubic interpolation, then immediately upsampled back to 128 x 128. This low-pass-filtered version serves as x_0 for the fine-scale DPIR solver. No denoising is performed at the coarse level. The purpose is to separate the effect of resolution cycling (which smooths the initialization) from the effect of actual coarse-level restoration.

3.3 Coarse PnP initialization

The experimental method. The observation is downsampled to 64 x 64. A complete PnP-DPIR solve is performed at this coarse resolution using a scaled blur kernel and the same DRUNet denoiser. The resulting coarse estimate is then upsampled to 128 x 128 via bicubic interpolation and used as x_0 for the fine-scale DPIR solver. The total runtime includes both the coarse and fine stages.

We deliberately kept everything else identical: same DRUNet weights, same HQS parameters, same number of fine-level iterations. The only variable is the starting point.

4. Evaluation Protocol

Reconstruction quality was measured using peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM), both computed on clipped [0, 1] RGB images against the original ground truth. Runtime was measured using CUDA-synchronized wall-clock timing on an NVIDIA Tesla T4 GPU.

Because random noise affects absolute PSNR values, we report paired differences: for each image-seed combination, we compute the PSNR difference between methods applied to the same degraded observation. This controls for noise variability and gives a more reliable estimate of the initialization effect.

5. Results

5.1 Overall performance

Table 1 summarizes the aggregate results across all 60 runs. The coarse PnP method achieves the highest mean PSNR and SSIM, followed by observation initialization, with direct DPIR as the baseline.

Method	Mean PSNR (dB)	Mean SSIM	Mean runtime (s)
Direct DPIR	28.411	0.8588	0.2719
Observation init	28.530	0.8603	0.2735
Coarse PnP init	**28.593**	**0.8622**	0.3772

Table 1. Overall results across 60 restoration experiments (4 images, 5 seeds, 3 methods). Boldface indicates the best value.

5.2 Paired comparisons

The paired analysis reveals that the coarse PnP initialization outperformed direct DPIR in every single one of the 20 image-seed combinations, with a mean gain of +0.182 dB. It also outperformed the observation-only initialization in all 20 cases, though by a smaller margin of +0.064 dB. The fact that observation initialization itself gains +0.118 dB over direct DPIR suggests that part of the benefit comes simply from smoothing the initial estimate. But the additional +0.064 dB from coarse PnP confirms that the coarse-level denoiser contributes something beyond low-pass filtering.

Comparison	Mean gain (dB)	Positive / Total
Coarse PnP vs Direct	+0.1820	20 / 20
Obs init vs Direct	+0.1183	20 / 20
Coarse PnP vs Obs init	+0.0637	20 / 20

Table 2. Paired PSNR differences. Every comparison is positive in all 20 trials.

5.3 Image-level breakdown

The gains are not uniform across images. Coffee, which has broad tonal areas and smooth gradients, benefits the most (+0.397 dB). Astronaut, with a mix of face detail and fabric texture, sees +0.287 dB. Chelsea (fine fur) and Rocket (sharp edges) show gains below +0.03 dB. This pattern makes physical sense: downsampling to 64 x 64 discards high-frequency content. Images whose structure is well-captured at low resolution benefit from the coarse stage; images dominated by fine detail do not.

Image	Coarse vs Direct	Coarse vs Obs init	Obs init vs Direct	Std. dev.
Coffee	+0.397 dB	+0.112 dB	+0.285 dB	0.052 dB
Astronaut	+0.287 dB	+0.112 dB	+0.175 dB	0.028 dB
Rocket	+0.029 dB	+0.024 dB	+0.005 dB	0.007 dB
Chelsea	+0.015 dB	+0.007 dB	+0.008 dB	0.013 dB

Table 3. Paired PSNR gains broken down by test image, averaged over 5 noise seeds.

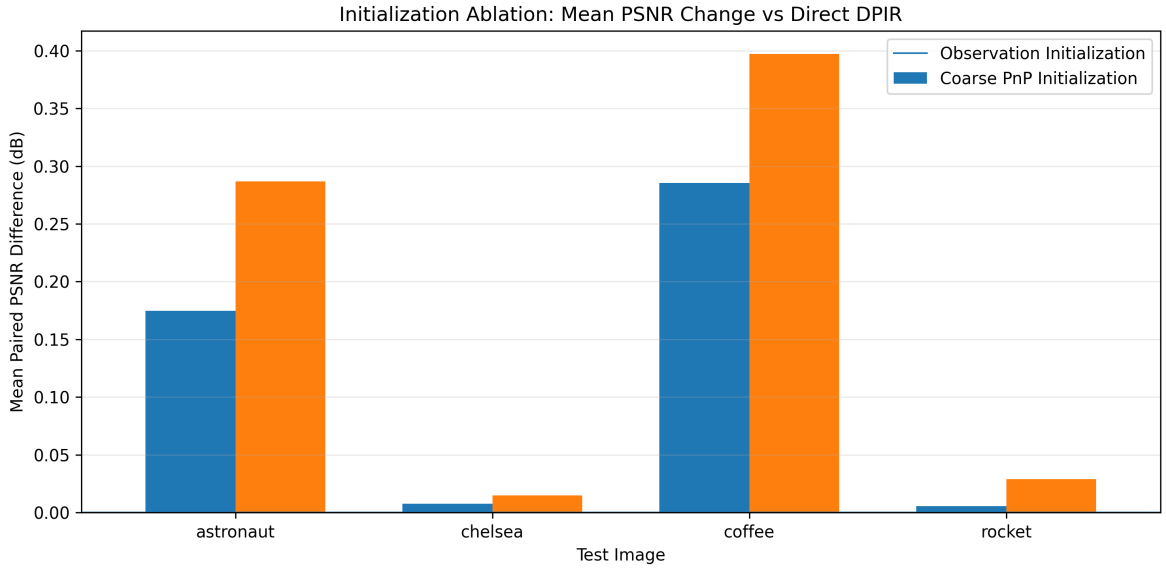


Figure 2. Per-seed PSNR comparison across the three initialization methods for each test image. The coarse PnP advantage is visible for Coffee and Astronaut but negligible for Chelsea and Rocket.

6. Baseline Comparison: TV-PGD vs Learned DPIR

Before investigating initialization strategies, we compared DRUNet-based DPIR against classical Total Variation (TV) regularization solved via proximal gradient descent. This provides context for the overall quality level of the learned prior. Table 4 shows the results on all four images at a single noise seed.

Image	Degraded	TV-PGD	DPIR	DPIR gain	TV time	DPIR time
Astronaut	19.18 dB	22.23 dB	24.13 dB	+1.90 dB	2.64 s	0.37 s
Chelsea	24.57 dB	27.22 dB	29.74 dB	+2.52 dB	2.66 s	0.35 s
Coffee	22.12 dB	25.95 dB	27.84 dB	+1.89 dB	2.35 s	0.37 s
Rocket	25.92 dB	29.15 dB	31.81 dB	+2.66 dB	2.61 s	0.36 s

Table 4. Classical TV-PGD vs learned DPIR. The learned denoiser prior produces +1.9 to +2.7 dB higher PSNR while running approximately 7 times faster.

The TV reconstructions exhibit the characteristic piecewise-constant staircasing artifacts, particularly visible on smooth gradients and skin tones. DPIR reconstructions are substantially sharper and more natural-looking. The TV-PGD regularization parameter was not exhaustively tuned; the comparison is meant to establish context rather than provide a definitive verdict on TV methods in general.

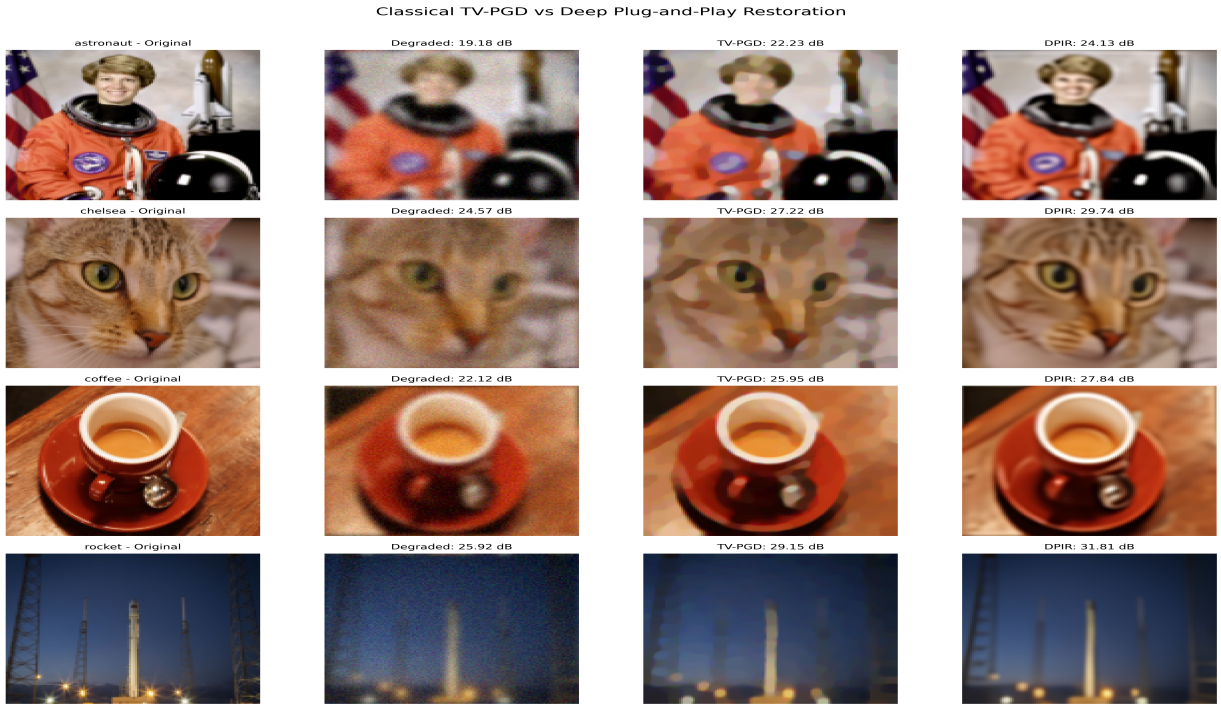


Figure 3. Visual comparison of ground truth, degraded observation, TV-PGD, and DRUNet DPIR across all four benchmark images.

7. Two-Level Restoration: Stage-by-Stage Visualization

Figure 4 shows the intermediate products of the coarse-to-fine pipeline. The coarse 64 x 64 restoration is visibly softer than the ground truth but captures overall structure, color distribution, and large-scale edges. After bicubic upsampling and the fine-scale DPIR pass, the output recovers substantially more detail. The visual improvement from coarse to fine is most apparent in textured regions.

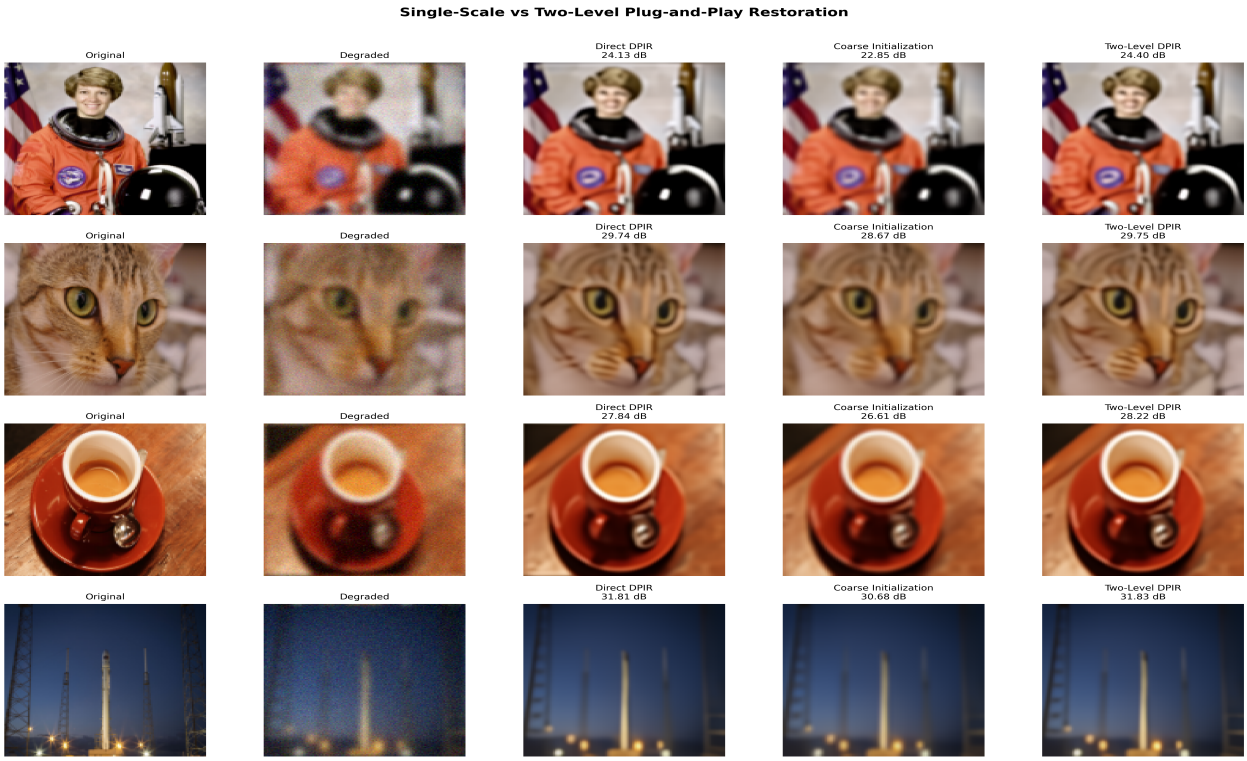


Figure 4. Multi-level restoration stages: coarse observation, coarse restored estimate (64 x 64), upsampled prolongation, and final refined output (128 x 128).

8. Convergence Behavior

We recorded the PSNR trajectory and relative residual at each HQS iteration. As shown in Figure 5, PSNR rises steeply during the first three to four iterations, then levels off. The relative residual (defined as the norm of the iterate difference divided by the current iterate norm) drops below 0.01 within five iterations. This confirms that the HQS solver converges reliably for this problem configuration, regardless of initialization strategy.

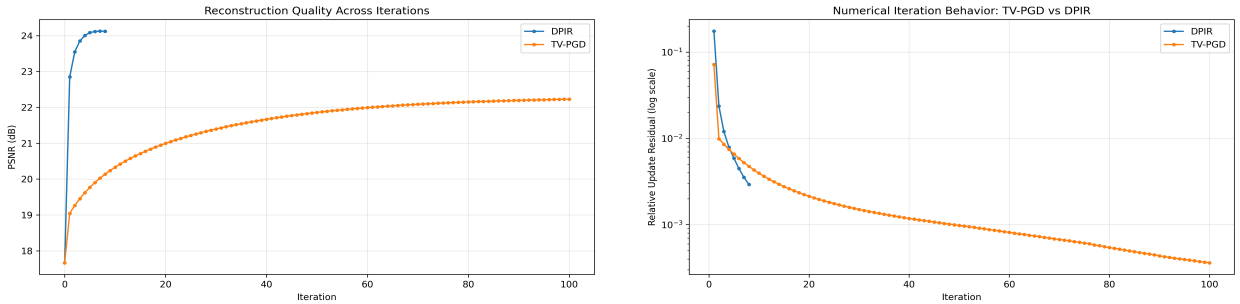


Figure 5. Left: PSNR progression over HQS iterations. Right: relative residual decay. Both confirm stable convergence within the allocated iteration budget.

9. Noise Robustness

To check whether the DRUNet prior remains effective under stronger noise, we repeated the single-image restoration experiment with noise standard deviations ranging from 0.01 (mild) to 0.07 (heavy). As expected, PSNR decreases monotonically with increasing noise. Importantly, the DRUNet denoiser degrades gracefully: there are no visible ringing artifacts or color shifts even at the highest noise level, though fine detail is progressively lost.

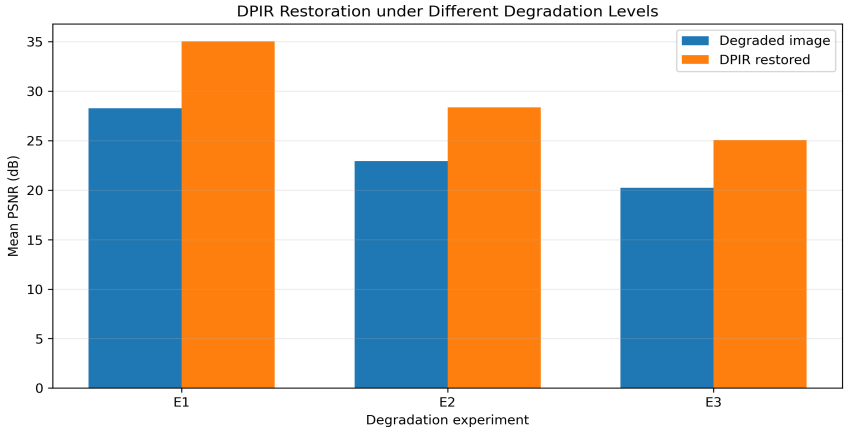


Figure 6. PSNR as a function of noise sigma. Performance degrades smoothly; no catastrophic failure is observed up to sigma = 0.07.

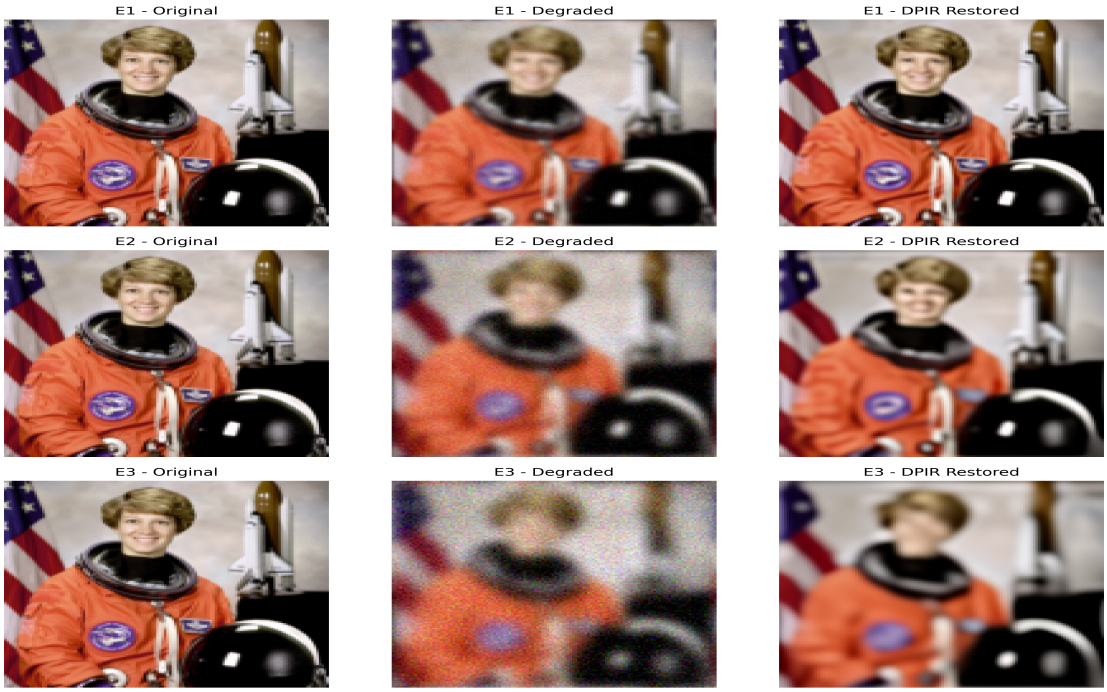


Figure 7. Visual restoration quality across noise levels (sigma = 0.01 to 0.07).

10. Discussion and Limitations

The paired results are clear: coarse PnP initialization produces higher PSNR than direct DPIR in every trial we ran. The effect is small in absolute terms (a fifth of a decibel on average) and varies substantially across images. But it is consistent, which suggests that the coarse estimate is providing genuinely useful structural information rather than just getting lucky on some noise realizations.

The observation initialization control is informative. It shows that roughly two-thirds of the total gain (+0.118 out of +0.182 dB) comes from smoothing the initialization via downsampling and upsampling. Only the remaining +0.064 dB can be attributed to the coarse DRUNet denoising step. This is a modest contribution, and it comes at the cost of a full additional PnP solve at coarse resolution.

The runtime numbers make the trade-off concrete. The coarse PnP method takes 0.377 seconds on average versus 0.272 seconds for direct DPIR, a 39% increase. Whether +0.18 dB justifies that overhead depends on the application. For offline processing of scientific imaging data, it might. For real-time video processing, it almost certainly would not.

There are several limitations that need to be stated plainly:

Small image sample. Four images and five noise seeds do not constitute a statistically powerful dataset. The per-image differences (Coffee +0.40 dB, Chelsea +0.01 dB) are observations about these specific images, not population-level estimates.

Approximate coarse operator. We constructed the 64 x 64 blur operator by resizing the fine-scale kernel, not by mathematically deriving a restriction operator. In a proper multigrid framework, the coarse operator should satisfy $A_c = R A P$, where R and P are restriction and prolongation operators. Our heuristic approach does not guarantee this consistency.

No flow-matching implementation. This study uses standard DRUNet denoising within an HQS loop. It does not implement flow-matching, score-based diffusion, or any generative restoration model. The work is related to but distinct from the Inria PnP-Flow project [4].

Single GPU, single run. All experiments were run on a single Tesla T4 GPU in Google Colab. The metrics were extracted from a single Colab session. While we verified file integrity and numerical consistency, a completely independent rerun in a fresh environment was not performed.

11. Reproducibility

The complete experiment is contained in a single Google Colab notebook available at notebooks/DeepInverse_PnP_Image_Restoration.ipynb. To reproduce the results: open the notebook in Colab, select a T4 GPU runtime, and run cells sequentially from top to bottom. The first cell installs DeepInverse 0.5.0; subsequent cells load pretrained DRUNet weights from the internet.

All numerical results are stored as CSV files in the results/ directory. Research figures are in figures/. Cryptographic SHA-256 checksums for all files are recorded in SHA256SUMS.txt. The exact software environment is documented in environment.json:

Component	Version
Python	3.13.16
PyTorch	2.11.0+cu130
DeepInverse	0.5.0
NumPy	2.1.3
Matplotlib	3.10.0
scikit-image	0.25.2
GPU	NVIDIA Tesla T4

Table 5. Recorded software and hardware environment.

12. Conclusion

We tested whether a two-level coarse-to-fine initialization helps PnP-DPIR image restoration. On four test images with five noise seeds each, it does: the mean paired PSNR gain is +0.182 dB, positive in all 20 trials. About two-thirds of this gain comes from smoothing the initialization; the remaining third comes from coarse-level denoising. The cost is a 39% increase in computation time.

The practical significance of these findings is limited. The image sample is small, the gains are modest, and the coarse operator is approximate. But the results are consistent and reproducible, and they suggest that multi-resolution initialization is worth exploring further, particularly for larger images and more challenging degradation models where the coarse stage might provide proportionally greater benefit.

References

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